

Parallel and Distributed Computing (CS3006)

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Course Instructors

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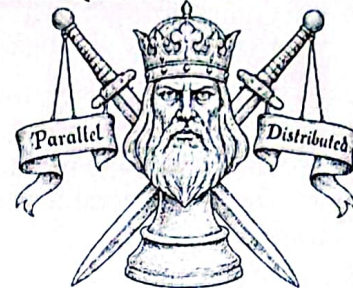
Dr. Abdul Qadeer

Final Exam

Total Time (Hrs.): 3

Total Marks: 80

Total Questions: 4



Roll No

Section

Student Signature

Instructions:

- This exam expects you to carefully think and then write your answer. Do not rush.
- Attempt all four questions. Solve all questions on the answer sheet.
- Marks per question and per sub-part are shown in square brackets []. Manage your time accordingly.
- If a question is ambiguous, note your assumption and proceed.
- Write legibly. Use the minimum number of words necessary. An unreadable answer earns no credit.

"The purpose of computing is insight, not numbers." — Richard Hamming

Good luck. Show your thinking, not just your answer.

Question	Marks	Time	Question	Marks	Time
Q1 — MPI Communication Cost Analysis	20	35 min	Q3 — MapReduce Pipeline Design	20	40 min
Q2 — Read, Diagnose and Fix: MPI C Code	20	45 min	Q4 — GFS Design Reasoning	20	40 min
Total	80	160 min			

* Suggested times total 160 minutes. The remaining 20 minutes are yours to read, think, and review — do not rush into writing.

 Recall & Apply [39 marks] Tests direct knowledge and application of concepts covered in lectures.	★ Reasoning & Judgment [41 marks] Requires analysis, design thinking, and reasoning from first principles.
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CLO 3: Analyze performance, scalability, and efficiency of parallel/distributed systems.

Q1: [20 marks] MPI Communication Cost Analysis — Suggested Time: 35 min

A climate simulation team distributes computation across 256 MPI processes arranged in an 8×32 two-dimensional mesh (8 rows, 32 columns). Each process holds one tile (a rectangular sub-region of the grid) of a large atmospheric grid. At each simulation step, every process must share its current boundary values with all other processes so a globally consistent pressure field can be computed.

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⦿ (a) State the following properties of an 8×32 2D mesh with 256 nodes: (i) diameter, (ii) arc-connectivity, and (iii) bisection width. Then identify which two of the three properties you stated above most directly constrain all-to-all broadcast performance and briefly state why. [5 marks]

⦿ (b) In the naive all-to-all approach where each of the 256 processes sends data directly to every other process: (i) how many messages does each process send? (ii) what is the total number of messages in the network? [5 marks]

★ (c) The team switches to a two-phase approach: first broadcast within each row, then broadcast across columns. Derive the number of communication steps for this two-phase approach for the 8×32 mesh. You may use the specific values for this 8×32 mesh or express your answer in general terms. Explain why this is more efficient than the naive approach. [5 marks]

★ (d) A colleague suggests replacing the 8×32 mesh with a hypercube of 256 processes. Name one property of the hypercube that would reduce communication cost for this workload, and one practical reason the team might still prefer the mesh. [5 marks]

CLO 4: Implement distributed applications handling concurrency and fault tolerance.

Q2: [20 marks] Read, Diagnose, and Fix — MPI C Code — Suggested Time: 45 min

Read the following MPI C code carefully.

```

1 #include <mpi.h>
2 #include <stdio.h>
3 #include <stdlib.h>
4
5 int main(int argc, char** argv) {
6     MPI_Init(&argc, &argv);
7     int rank, size;
8     MPI_Comm_rank(MPI_COMM_WORLD, &rank);
9     MPI_Comm_size(MPI_COMM_WORLD, &size); // size = 4
10    int N = 1000, blk = N / size; // partition into 4
11    int buf[blk], partial = 0; // chunk=250
12    if (rank == 0) {
13        int data[1000];
14        for (int i = 0; i < N; i++) data[i] = i + 1;
15        for (int i = 1; i < size; i++)
16            MPI_Send(data + i*blk, blk, MPI_INT, i, 0, MPI_COMM_WORLD);
17        for (int j = 0; j < blk; j++) partial += data[j];
18        MPI_Ssend(&partial, 1, MPI_INT, 0, 1, MPI_COMM_WORLD);
19    }
20    if (rank != 0) {
21        MPI_Recv(buf, blk, MPI_INT, 0, 0, MPI_COMM_WORLD, MPI_STATUS_IGNORE);
22        for (int j = 0; j < blk; j++) partial += buf[j];
23        MPI_Send(&partial, 1, MPI_INT, 0, 1, MPI_COMM_WORLD);
24    }
25    if (rank == 0) {
26        int result = 0, val;
27        for (int i = 0; i < size; i++) {
28            MPI_Recv(&val, 1, MPI_INT, i, 1, MPI_COMM_WORLD, MPI_STATUS_IGNORE);
29            result += val;
30        }
31        printf("%d\n", result);
32    }
33    MPI_Finalize();

```

Handwritten notes:

- At line 10: "partition into 4" with an arrow pointing to `blk = N / size`.
- At line 11: "chunk=250" with an arrow pointing to `blk`.
- At line 16: "Populating array [1...1000]" with an arrow pointing to the `data` array initialization.
- At line 18: "dest rank" with an arrow pointing to the `0` in the `MPI_Ssend` call.
- At line 21: "buf [250] src rank" with an arrow pointing to the `0` in the `MPI_Recv` call.
- At line 28: "src rank" with an arrow pointing to the `i` in the `MPI_Recv` call.

```
return 0;
```

- ✱ (a) What does this program intend to do? Describe the computational goal in two sentences without referring to any MPI function names. [3 marks]
- ★ (b) There is a critical bug in this code. Identify it precisely: (i) which lines are responsible, (ii) what happens at runtime, and (iii) what is the technical term for this class of error in MPI programming? [7 marks]
- ★ (c) Fix the bug. Rewrite only the affected lines in C. Explain in one sentence why your fix resolves the problem. [5 marks]
- ✱ (d) This code uses point-to-point communication for two distinct communication phases. Replace each phase with the appropriate MPI collective operation. For each, write the function name and its key arguments, and briefly state what it does in this context. Exact argument order will not be penalised. [5 marks]

CLO 2: Design parallel algorithms and distributed solutions using modern programming models.

Q3: [20 marks] MapReduce Pipeline Design — Suggested Time: 40 min

A public health agency maintains 2 billion anonymized medical records, each containing: patient_id, city, age, diagnosis_code, and visit_date. The dataset totals approximately 4TB across hundreds of files. The agency runs the following analysis monthly:

For each diagnosis code, compute the average patient age — but only for cities where that diagnosis code appears more than 10,000 times in the last 12 months.

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- ✱ (a) Describe MapReduce: (i) name the two main computation phases, (ii) describe in 2-3 sentences what happens in each phase, (iii) what happens to intermediate key-value pairs between the two phases? [5 marks]
- ★ (b) Design a MapReduce job for the monthly analysis described above. State precisely: what a single mapper receives as input and what key-value pair(s) it emits; what the reducer receives and what it outputs. One sentence each is sufficient if precise. [7 marks]
- ✱ (c) Diagnosis code Z99 appears in 800 million of the 2 billion records. (i) Name the MapReduce performance problem this situation creates. (ii) In 2 sentences explain why it degrades performance. [4 marks]
- ★ (d) One worker machine crashes mid-job. Since a worker has multiple cores and each core was executing an independent map task, several map tasks are lost simultaneously. What does the MapReduce framework automatically do in response, and why does this not corrupt the final output? Reference a specific design property of MapReduce. [4 marks]

CLO 1: Explain the architectures and principles of parallel and distributed computing.

Q4: [20 marks] GFS Design Reasoning — Suggested Time: 40 min

Answer the following questions about the design and behaviour of the Google File System (GFS).

- ⚙ (a) Name the three main components of GFS and describe the role of each in 1-2 sentences. **[5 marks]**
- ⚙ (b) A research team stores a 192MB file in GFS with 64MB chunks and 3-way replication. (i) How many chunks does this file require? (ii) How many total chunk replicas are stored? (iii) Which GFS component decides which chunkservers hold each replica? **[2 marks]**
- ★ (c) Many clients wrote a total of 192MB of data to a GFS file using atomic record append. Will the actual size of this file in GFS be exactly 192MB, less, or potentially more than 192MB? Justify your answer by referencing how GFS handles record append failures and what the consistency model says about the file content after concurrent appends. **[3 marks]**
- ★ (d) A web application stores millions of user profile pictures, each between 50KB and 200KB, written once and read frequently. Identify two specific GFS design decisions that make it a poor fit for this workload. Explain each in 2-3 sentences. **[5 marks]**
- ⚙ (e) The GFS master stores all filesystem metadata in memory. A new engineer argues this is reckless — a single master crash would lose all metadata. How does GFS address this concern? Is the engineer's worry completely unfounded? Explain in 3-4 sentences. **[5 marks]**

— END OF EXAMINATION —